

Inmon Vs Kimball approaches

Inmon Approach (Top-Down)

advantage

- Flexibility. Inmon's approach is much more flexible to changing business requirements and more flexible to modifications to the data sources
- Single source of truth. Because of the normalized data model, the data warehouse acts as a single source of truth throughout the enterprise
- Less prone to errors. Normalization avoids data redundancy so both the engineering processes and analytic processes are less prone to errors
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Disadvantage

- Initial set-up and maintenance cost. Setting up and maintaining Inmon's architecture is much more expensive
- Skills needed. Inmon's approach necessitates highly skilled engineers

Kimball

advantage

1. Simplicity and speed. Kimball's architecture is much simpler and faster to design and set up.
2. Understandable. The dimensional data model is easy to understand for non-technical and technical employees alike
3. Engineering team needs. Kimball's approach needs fewer engineers with less specialized technical skills to both set up and maintain the data warehouse than Inmon's approach.

Disadvantages

1. Data redundancy. Because data is loaded into a dimensional model, there is more data redundancy and therefore a higher chance for errors.
- 2-No single source of truth. Data within the data warehouse is designed and organized around data marts. Coupled with data redundancy,
Less flexible.

2-Full Load Vs Incremental Load

Full Load overwrites all data in the target, best for smaller datasets or initial setups, ensuring uniformity but high resource consumption.
Incremental Load syncs only new or changed data, ideal for large datasets or dynamic environments, enhancing efficiency but requiring complex change tracking mechanisms

